Coral Trout Abundance

Count models and overdispersion

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Study and findings

This observational analysis uses a supplied teaching extract containing 120 anonymised Great Barrier Reef monitoring sites from the 2012 survey. Each site count sums five equal-length transects. The response combines recorded Plectropomus species.

Substantial overdispersion (Pearson dispersion 5.38) motivates a quasi-Poisson count model. After accounting for this variation, the latitude-by-zone interaction is not supported (F-test p = 0.420), and an additive model is retained.

Conditional on latitude band, No-take sites have an estimated expected-count ratio of 1.76 relative to Open sites. The approximate 95% interval is 1.24 to 2.49, calculated using plus/minus two standard errors on the log scale. This association does not establish a causal effect of zoning.

Data acknowledgement

Data were sourced from the Australian Institute of Marine Science (AIMS) Long-Term Monitoring Program visual-census fish dataset. The course teaching extract anonymises site identifiers and defines broad latitude categories. These categories are not official AIMS management regions.

Source metadata: https://apps.aims.gov.au/metadata/view/5be0b340-4ade-11dc-8f56-00008a07204e

About this public copy

The following pages retain the original completed analysis, code, tables and figures. Course instructions and identifying submission headers have been omitted. Model calculations have been reproduced in R 4.5.2. The source files remain unchanged.

Code, data provenance and recomputed outputs: https://github.com/xialuote77/coral-trout-glm-analysis

Part 1: Executive summary

This report examined how coral trout abundance varied across Northern, Central and Southern latitude bands and between reefs that were open to fishing and those in No-take zones in the 2012 survey. The analysis used a count model that allowed for substantially greater variability in coral trout numbers than would be expected under a standard Poisson model.

Coral trout abundance was much higher in the Southern band than in the Northern and Central bands. Fitted site-level mean counts were 2-4 fish in the Northern and Central bands, compared with about 8.5 in Southern Open sites and 14.9 in Southern No-take sites. After accounting for latitude, sites in No-take zones were estimated to have 1.76 times the expected coral trout count of sites open to fishing, with a 95% confidence interval from 1.24 to 2.49.

Although the initial analysis suggested that the association with fishing zone might differ across latitude bands, this was no longer supported after allowing for substantial extra variability in the counts. Model diagnostics identified a small number of unusual and influential sites, particularly AIMS022 and AIMS024. However, sensitivity analyses showed that removing these sites reduced the estimated magnitude of some effects but did not change the main conclusions. The most important limitation is that the data are observational, so the observed differences between fishing zones should be interpreted as associations rather than evidence that zoning itself caused the differences in coral trout abundance.

Part 2: Problem and data understanding

```
reef <- read.csv("aims_coral_trout_sites_2012.csv")

reef$latitude_band <- factor(
  reef$latitude_band,
  levels = c("Northern", "Central", "Southern")
)
reef$zone <- factor(
  reef$zone,
  levels = c("Open", "No-take")
)
```

```r
dim(reef)
```

```
[1] 120   4
```

```r
names(reef)
```

```
[1] "site"          "latitude_band" "zone"          "coral_trout"
```

```r
str(reef)
```

```
'data.frame':   120 obs. of  4 variables:
 $ site         : chr  "AIMS001" "AIMS002" "AIMS003" "AIMS004" ...
 $ latitude_band: Factor w/ 3 levels "Northern","Central",..: 3 3 3 3 3 3 3 3 3 3 ...
 $ zone         : Factor w/ 2 levels "Open","No-take": 2 2 2 2 2 2 2 2 2 2 ...
 $ coral_trout  : int  5 9 3 13 0 5 13 19 5 24 ...
```

```r
summary(reef)
```

```
       site      latitude_band      zone     coral_trout
 Length   :120   Northern:36    Open   :60   Min.   : 0.000
 N.unique :120   Central :36    No-take:60   1st Qu.: 1.000
 N.blank  :  0   Southern:48                 Median : 4.000
 Min.nchar:  7                              Mean   : 6.417
 Max.nchar:  7                              3rd Qu.: 7.000
                                            Max.   :61.000
```

```r
colSums(is.na(reef))
```

```
        site latitude_band          zone   coral_trout
           0             0             0             0
```

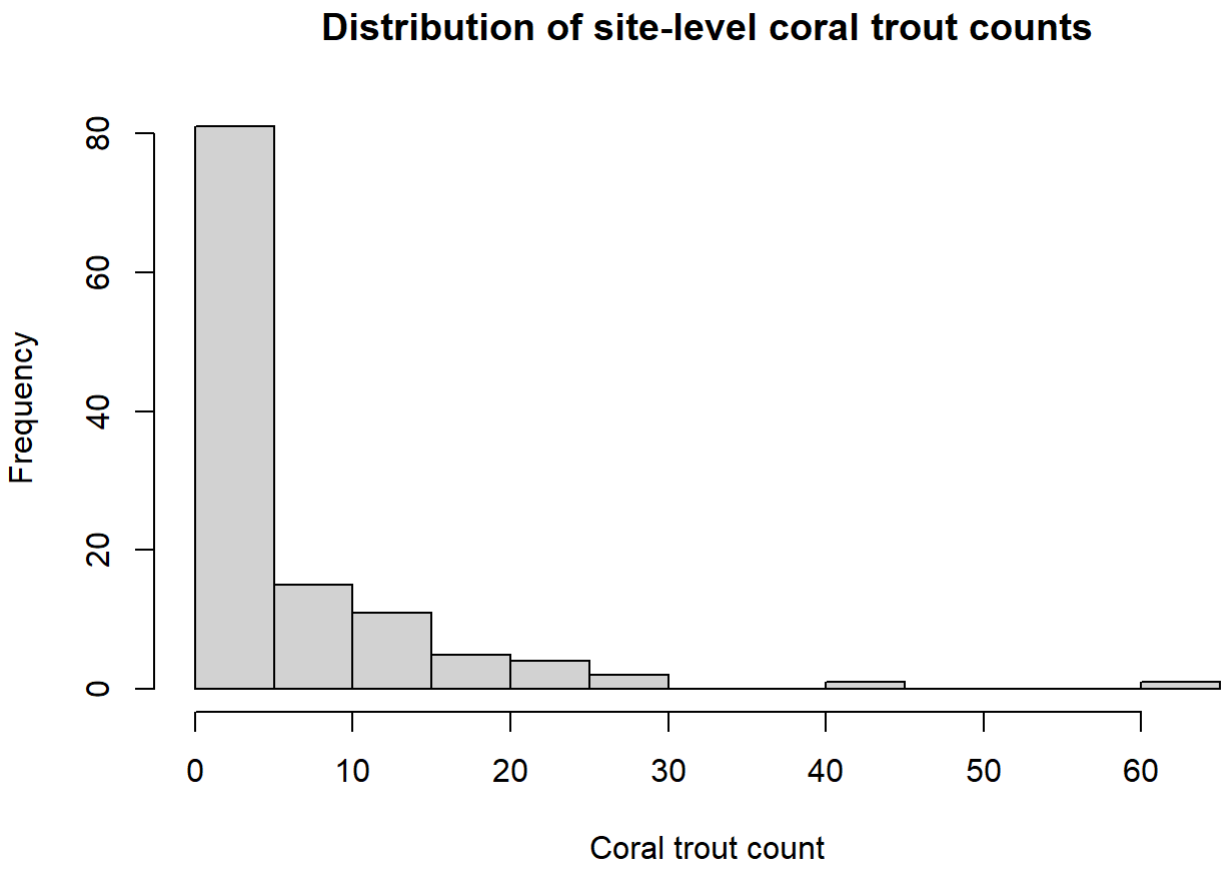
The statistical objective is to assess whether expected site-level coral trout abundance is associated with latitude band and fishing zone, and whether the association with fishing zone differs among latitude bands. The response variable is `coral_trout`, the total number of coral trout recorded across five equal-length transects at each monitoring site. The explanatory variables are `latitude_band` (Northern, Central, Southern) and `zone` (Open, No-take). The observational unit is a monitoring site, with each row representing one site in the 2012 survey. As the study is observational rather than experimental, any association with fishing zone should not be interpreted as causal.

The dataset contains 120 sites and four variables. `site` was stored as a character variable, `latitude_band` and `zone` as factors, and `coral_trout` as an integer-valued count. Northern and Open were used as the reference levels. There were 36 Northern, 36 Central and 48 Southern sites, with 60 Open and 60 No-take sites. `coral_trout` ranged from 0 to 61. No missing values were present in any of the four variables.

Part 3: Exploratory analysis

Exploratory analysis was then used to examine the count distribution and possible differences across latitude bands and zones.

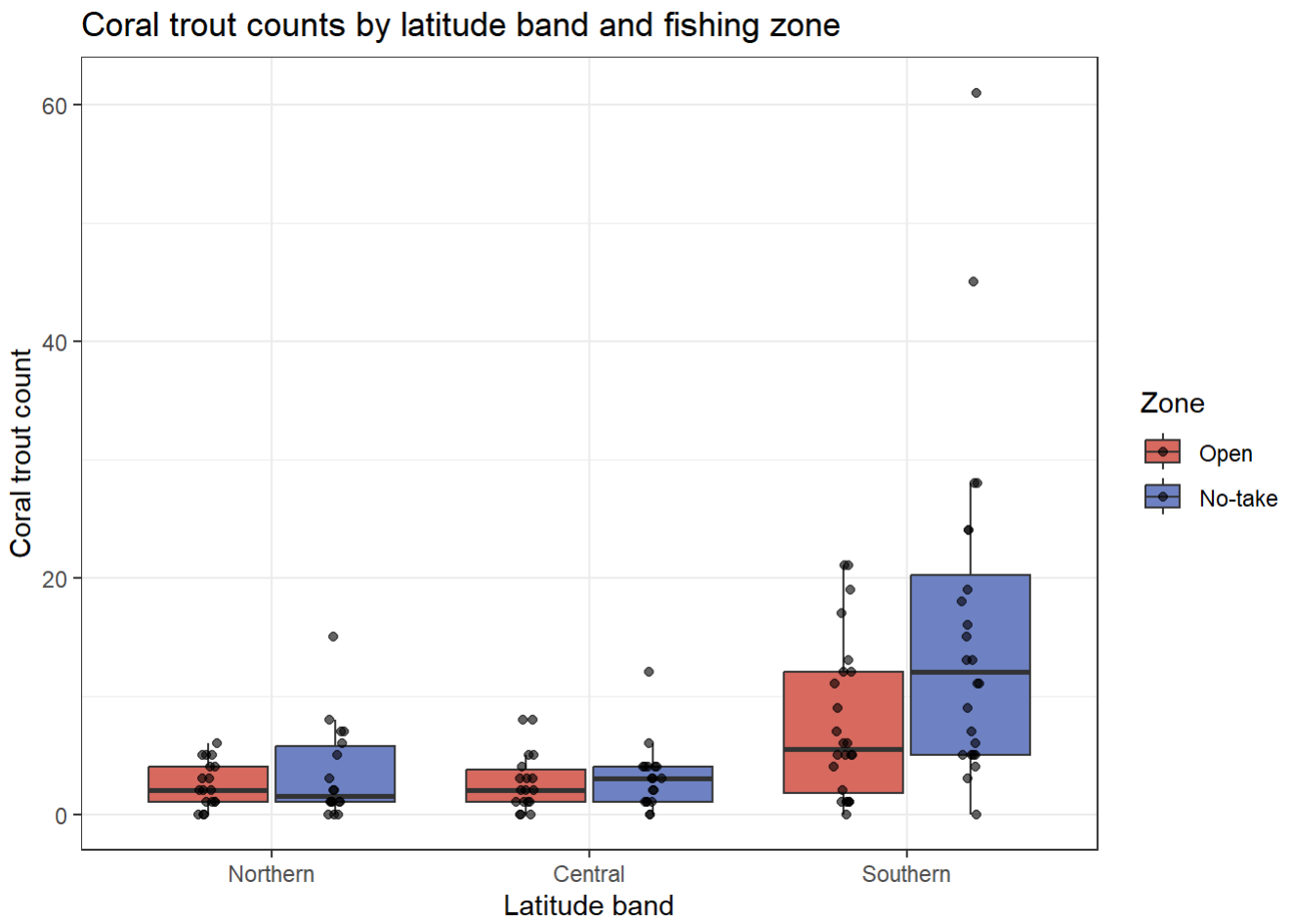
```r
hist(reef$coral_trout,
     breaks = 15,
     xlab = "Coral trout count",
     main = "Distribution of site-level coral trout counts")
```



The histogram shows a strongly right-skewed distribution. Most sites recorded relatively few coral trout, while a small number of sites had much larger counts, producing a long right tail. This substantial variability motivates careful assessment of the variance assumption in the subsequent count model.

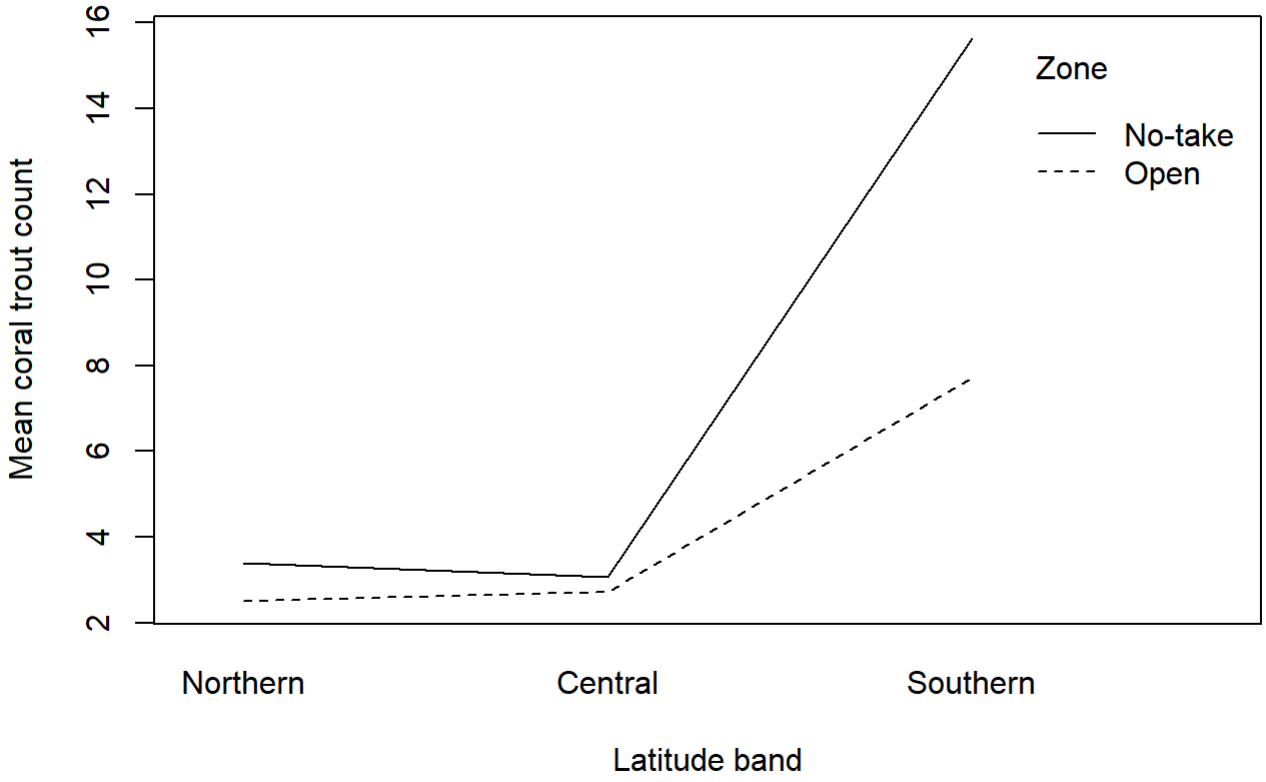
```r
library(ggplot2)

ggplot(reef, aes(x = latitude_band, y = coral_trout, fill = zone)) +
  geom_boxplot(outlier.shape = NA,
               position = position_dodge(width = 0.8)) +
  geom_point(aes(group = zone),
             position = position_jitterdodge(jitter.width = 0.12,
                                             dodge.width = 0.8),
             size = 1.5,
             alpha = 0.6) +
  labs(x = "Latitude band",
       y = "Coral trout count",
       fill = "Zone",
       title = "Coral trout counts by latitude band and fishing zone") +
  theme_bw()+
  scale_fill_manual(values = c("Open" = "#D96C5F",
                               "No-take" = "#7183C7"))
```



Differences were also apparent across latitude bands. Northern and Central sites generally had low coral trout counts with substantial overlap between Open and No-take zones, whereas Southern sites had higher counts and greater variability. Within the Southern band, No-take sites tended to have higher counts than Open sites. Several unusually large observations were also present, particularly among Southern No-take sites, and these will be examined later using model diagnostics.

```
with(reef,
     interaction.plot(latitude_band, zone, coral_trout,
                      xlab = "Latitude band",
                      ylab = "Mean coral trout count",
                      trace.label = "Zone"))
```



Consistent with this pattern, the interaction plot suggests that the association between fishing zone and coral trout abundance may differ across latitude bands. Mean counts for Open and No-take sites were relatively similar in the Northern and Central bands, but the difference became much larger in the Southern band. The non-parallel profiles therefore provide preliminary evidence of an interaction, which will be formally assessed in the GLM analysis.

```
group_summary <- aggregate(
  coral_trout ~ latitude_band + zone,
  data = reef,
  FUN = function(x) c(n = length(x),
                      mean = mean(x),
                      median = median(x),
                      variance = var(x))
  )
group_summary <- cbind(group_summary[c("latitude_band", "zone")],
                       as.data.frame(group_summary$coral_trout))

names(group_summary) <- c("Latitude band", "Zone", "n", "Mean", "Median", "Variance")

group_summary <- group_summary[order(group_summary$`Latitude band`, group_summary$Zone),]

knitr::kable(group_summary, digits = 2,
             caption = "Summary of coral trout by latitude band and fishing zone")
```

Summary of coral trout by latitude band and fishing zone

	Latitude band	Zone	n	Mean	Median	Variance
1	Northern	Open	18	2.50	2.0	3.79
4	Northern	No-take	18	3.39	1.5	15.55
2	Central	Open	18	2.72	2.0	6.09
5	Central	No-take	18	3.06	3.0	7.70
3	Southern	Open	24	7.71	5.5	43.95
6	Southern	No-take	24	15.62	12.0	201.90

The numerical summaries support these visual patterns. Mean counts were similar in Northern and Central sites, ranging from 2.50 to 3.39 across zones, but were much higher in the Southern band. In Southern sites, the mean was 7.71 in Open zones and 15.63 in No-take zones. Variability was also greatest in the Southern groups, particularly for No-take sites.

Part 4: Initial GLM choice and formulation

These patterns motivate an initial count GLM including latitude band, zone and their interaction.

Let Y_i denote the total number of coral trout observed across the five transects at site i, and let $\mu_i = E(Y_i)$. As the response is a non-negative integer count recorded under the same sampling effort of five equal-length transects at each site, a Poisson model provides a natural initial model for the site-level counts. A log link was used so that fitted expected counts remain positive:

$$Y_i \sim \text{Poisson}(\mu_i), \qquad E(Y_i) = \mu_i, \qquad \text{Var}(Y_i) = \mu_i$$

Site observations are assumed to be independent, conditional on latitude band and zone. Using Northern and Open as the reference levels,

$$\log(\mu_i) = \beta_0 + \beta_1 C_i + \beta_2 S_i + \beta_3 N_i + \beta_4 C_i N_i + \beta_5 S_i N_i,$$

where C_i, S_i, and N_i indicate Central, Southern, and No-take sites respectively. The log link ensures $\mu_i = \exp(\eta_i) > 0$, so fitted expected counts are always valid.

```
poisson_int <- glm(coral_trout ~ latitude_band * zone,
                   family = poisson(link = "log"),
                   data = reef)

summary(poisson_int)
```

```
Call:
glm(formula = coral_trout ~ latitude_band * zone, family = poisson(link = "log"),
    data = reef)

Coefficients:
                                Estimate Std. Error z value Pr(>|z|)
(Intercept)                      0.91629    0.14907   6.147 7.91e-10 ***
latitude_bandCentral             0.08516    0.20647   0.412   0.6800
latitude_bandSouthern            1.12601    0.16622   6.774 1.25e-11 ***
zoneNo-take                      0.30421    0.19651   1.548   0.1216
```

```
latitude_bandCentral:zoneNo-take  -0.18870    0.27786  -0.679   0.4971
latitude_bandSouthern:zoneNo-take  0.40236    0.21607   1.862   0.0626 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 977.60  on 119  degrees of freedom
Residual deviance: 570.62  on 114  degrees of freedom
AIC: 939.3

Number of Fisher Scoring iterations: 5
```

With Northern and Open as the reference levels, the intercept corresponds to an expected count of $e^{0.916} = 2.50$ coral trout for a Northern Open site. Among Open sites, the expected count was about 1.09 times as large in Central sites and 3.08 times as large in Southern sites as in Northern sites. Within the Northern band, No-take sites had an expected count about 1.36 times that of Open sites. The interaction terms modify this zone association in the other latitude bands: the No-take/Open ratio was multiplied by about 0.83 in Central sites and 1.50 in Southern sites relative to Northern sites. Thus, the fitted model allows the association between fishing zone and expected coral trout abundance to vary by latitude band.

Part 5: Initial model comparison

To assess whether the association between fishing zone and expected coral trout abundance differed across latitude bands, the additive and interaction Poisson models were compared. The null hypothesis was that both interaction coefficients were zero,

$$H_0 : \beta_4 = \beta_5 = 0$$

corresponding to the same No-take/Open association across latitude bands. The alternative hypothesis was that at least one interaction coefficient was non-zero,

$$H_A : \text{at least one of} \beta_4, \beta_5 \neq 0.$$

```r
poisson_add <- glm(coral_trout ~ latitude_band + zone,
                   family = poisson(link = "log"),
                   data = reef)

anova(poisson_add, poisson_int, test = "Chisq")
```

```
Analysis of Deviance Table

Model 1: coral_trout ~ latitude_band + zone
Model 2: coral_trout ~ latitude_band * zone
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1       116     580.03
2       114     570.62  2   9.4029 0.009082 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
AIC(poisson_add, poisson_int)
```

```
            df      AIC
poisson_add  4 944.7003
poisson_int  6 939.2974
```

The likelihood-ratio test gave a deviance difference of 9.40 on 2 degrees of freedom ($p = 0.0091$). Thus, under the initial Poisson model, there was evidence against H_0, suggesting that the association between fishing zone and expected coral trout abundance differed across latitude bands. The interaction model also had a lower AIC than the additive model (939.30 vs. 944.70), providing further support for the interaction structure. Therefore, the interaction model is provisionally retained, subject to checking the Poisson variance assumption in the next stage.

Part 6: Dispersion assessment

```r
(pearson_X2 <- sum(residuals(poisson_int, type = "pearson")^2))
```

```
[1] 613.0427
```

```r
(df_resid <- df.residual(poisson_int))
```

```
[1] 114
```

```r
(phi_hat <- pearson_X2 / df_resid)
```

```
[1] 5.377568
```

```r
(p_gof <- pchisq(deviance(poisson_int), df = df_resid, lower.tail = FALSE))
```

```
[1] 6.781645e-62
```

The Pearson statistic was $X^2 = 613.04$ on 114 residual degrees of freedom, giving a dispersion estimate of

$$\hat{\phi} = \frac{613.04}{114} = 5.38.$$

This is substantially greater than 1, indicating strong overdispersion relative to the Poisson assumption $\mathrm{Var}(Y_i) = \mu_i$. The residual deviance was 570.62 on 114 degrees of freedom, with a goodness-of-fit p-value of 6.78×10^{-62}, providing further evidence that the Poisson variance assumption is inadequate. Consequently, Poisson standard errors are likely to be underestimated and the inferential results from the initial model should be treated cautiously. An overdispersion model is therefore required.

Part 7: Overdispersion model and final selection

```r
quasi_add <- glm(coral_trout ~ latitude_band + zone,
                 family = quasipoisson(link = "log"),
                 data = reef)

quasi_int <- glm(coral_trout ~ latitude_band * zone,
                 family = quasipoisson(link = "log"),
```

```
                data = reef)

summary(quasi_int)
```

```
Call:
glm(formula = coral_trout ~ latitude_band * zone, family = quasipoisson(link = "log"),
    data = reef)

Coefficients:
                                  Estimate Std. Error t value Pr(>|t|)
(Intercept)                        0.91629    0.34569   2.651  0.00918 **
latitude_bandCentral               0.08516    0.47880   0.178  0.85915
latitude_bandSouthern              1.12601    0.38545   2.921  0.00420 **
zoneNo-take                        0.30421    0.45569   0.668  0.50575
latitude_bandCentral:zoneNo-take  -0.18870    0.64434  -0.293  0.77017
latitude_bandSouthern:zoneNo-take  0.40236    0.50106   0.803  0.42364
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasipoisson family taken to be 5.377587)

    Null deviance: 977.60  on 119  degrees of freedom
Residual deviance: 570.62  on 114  degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 5
```

```
anova(quasi_add, quasi_int, test = "F")
```

```
Analysis of Deviance Table

Model 1: coral_trout ~ latitude_band + zone
Model 2: coral_trout ~ latitude_band * zone
  Resid. Df Resid. Dev Df Deviance      F Pr(>F)
1       116     580.03
2       114     570.62  2   9.4029 0.8743 0.4199
```

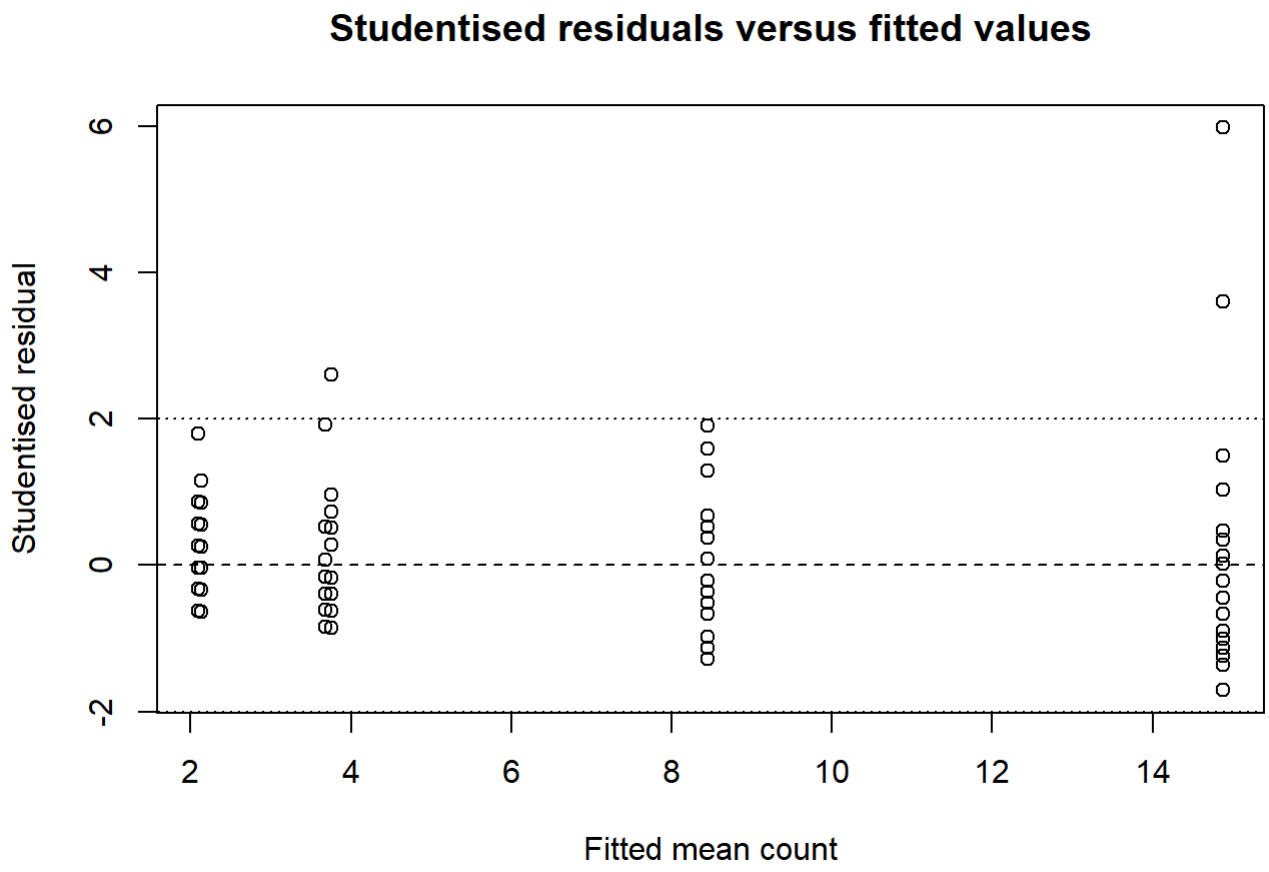
Because the Poisson model showed substantial overdispersion, the models were refitted using a quasi-Poisson distribution with a log link, allowing $\mathrm{Var}(Y_i) = \phi\mu_i$. The estimated dispersion parameter for the interaction model was $\hat{\phi} = 5.38$, which increased the standard errors relative to the original Poisson model while leaving the fitted mean structure unchanged. The additive and interaction quasi-Poisson models were compared using an F test, giving $F = 0.874$ and $p = 0.420$. After accounting for overdispersion, there was therefore insufficient evidence that the association between fishing zone and coral trout abundance differed across latitude bands. The simpler additive quasi-Poisson model was selected as the final model. AIC was not used because quasi-Poisson models are based on quasi-likelihood rather than a fully specified likelihood.

Part 8: Diagnostics and model alternatives

```
final_mod <- quasi_add
```

```r
plot(fitted(final_mod),
     rstudent(final_mod),
     xlab = "Fitted mean count",
     ylab = "Studentised residual",
     main = "Studentised residuals versus fitted values")

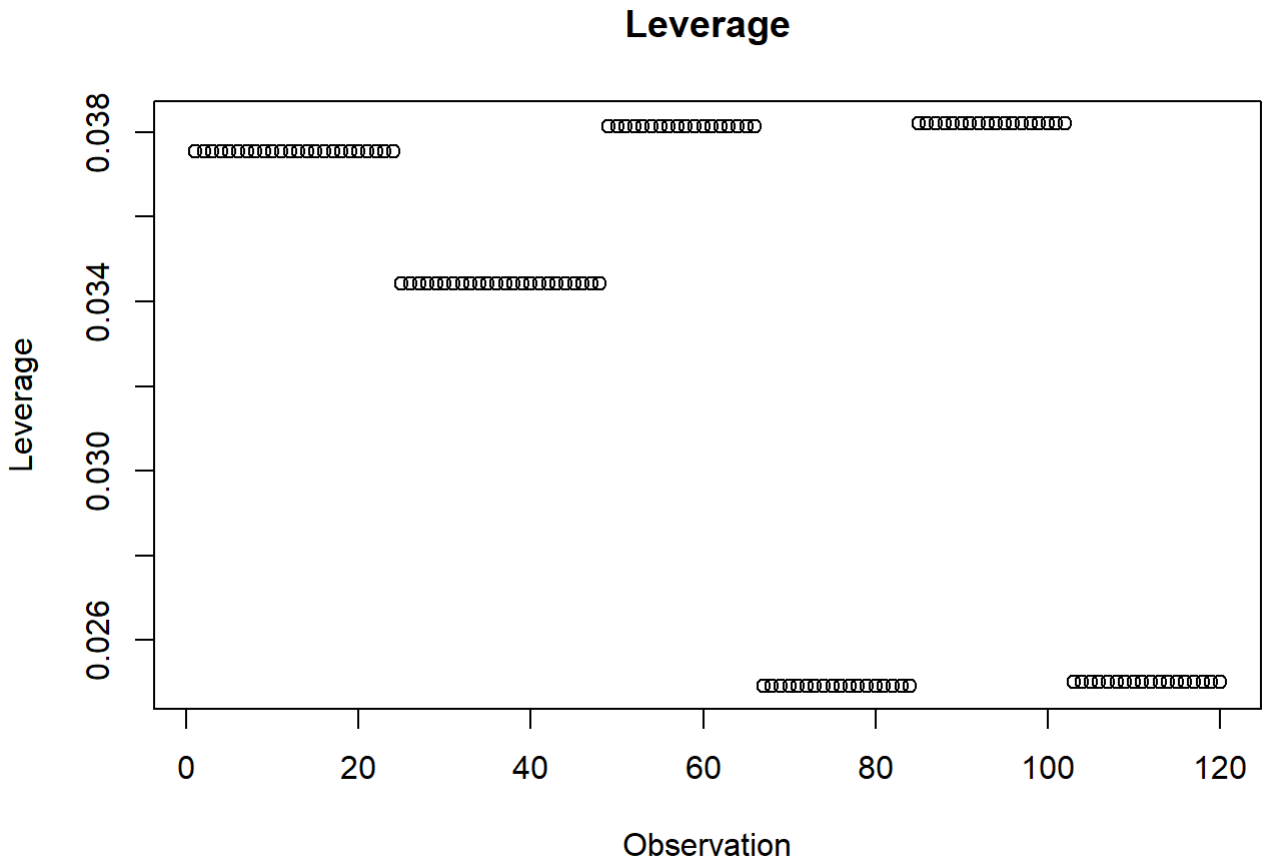
abline(h = 0, lty = 2)
abline(h = c(-2, 2), lty = 3)
```



The studentised residuals showed no clear systematic trend or curvature across fitted values, suggesting that the additive mean structure was broadly reasonable. Most residuals were within approximately ± 2, although AIMS022, AIMS024 and AIMS099 had unusually large positive residuals of 5.98, 3.60 and 2.61 respectively.

```r
lev <- hatvalues(final_mod)

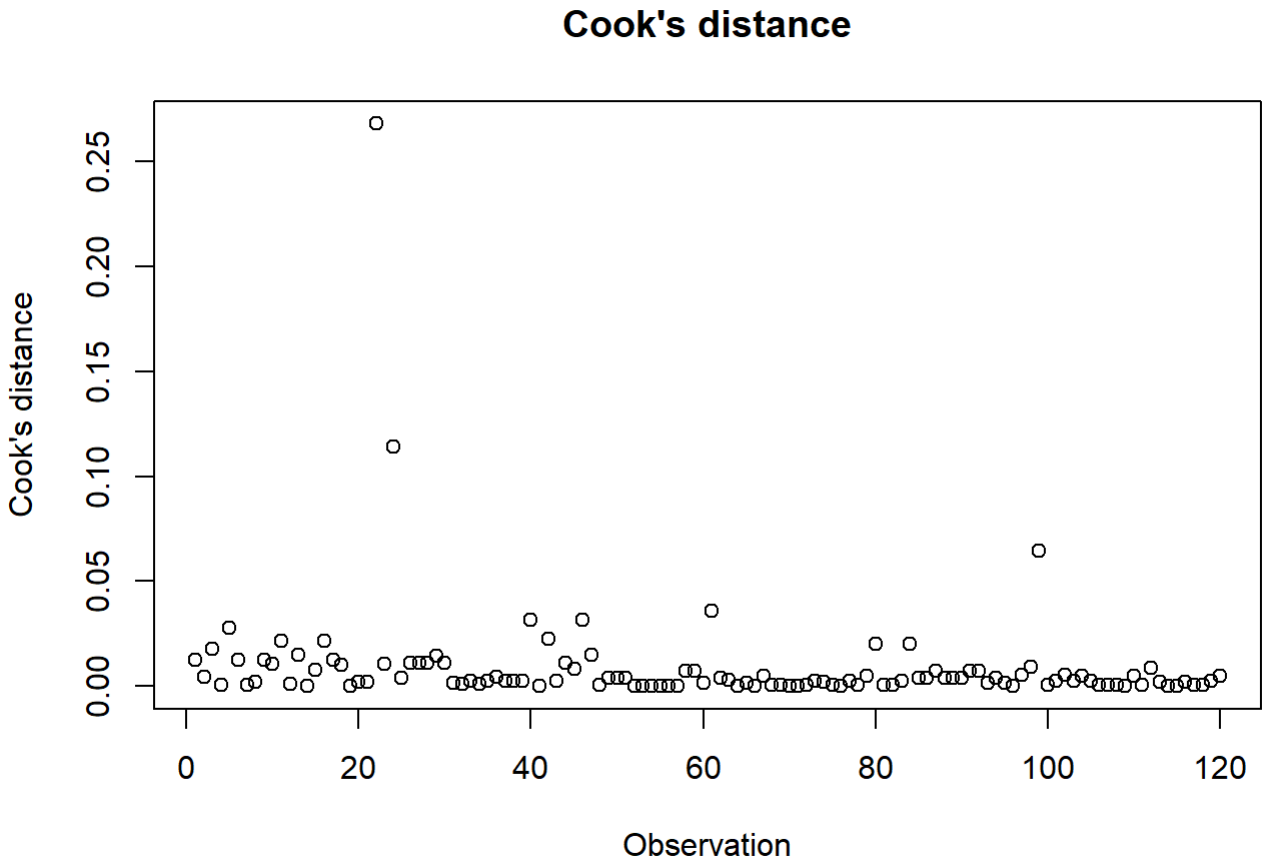
plot(lev, ylab = "Leverage", xlab = "Observation", main = "Leverage")
```



Leverage values were relatively similar across sites, with no observation showing exceptionally high leverage. This suggests that the unusual residuals were not primarily caused by sites having extreme predictor combinations.

```
cd <- cooks.distance(final_mod)

plot(cd, ylab = "Cook's distance", xlab = "Observation", main = "Cook's distance")
```



Cook's distance identified AIMS022 as the most influential observation, with AIMS024 also standing out. This motivated a sensitivity analysis to assess whether the final conclusions depended heavily on these sites.

```
diag_table <- data.frame(site = reef$site,
                         fitted = fitted(final_mod),
                         studentised_residual = rstudent(final_mod),
                         leverage = lev,
                         cooks_distance = cd)

diag_table[order(-abs(diag_table$studentised_residual)), ][1:10, ]
```

```
      site    fitted studentised_residual   leverage cooks_distance
22 AIMS022 14.878788             5.975892 0.03754918     0.26808934
24 AIMS024 14.878788             3.595725 0.03754918     0.11434654
99 AIMS099  3.755123             2.608132 0.03819683     0.06431947
61 AIMS061  3.684271             1.922150 0.03814453     0.03579842
40 AIMS040  8.454545             1.910211 0.03442049     0.03179258
46 AIMS046  8.454545             1.910211 0.03442049     0.03179258
80 AIMS080  2.093506             1.795247 0.02491463     0.02020025
84 AIMS084  2.093506             1.795247 0.02491463     0.02020025
5  AIMS005 14.878788            -1.705169 0.03754918     0.02790058
42 AIMS042  8.454545             1.598261 0.03442049     0.02246382
```

```
diag_table[order(-diag_table$cooks_distance), ][1:10, ]
```

```
      site    fitted studentised_residual   leverage cooks_distance
22 AIMS022 14.878788             5.975892 0.03754918     0.26808934
```

```
24 AIMS024 14.878788             3.595725 0.03754918     0.11434654
99 AIMS099  3.755123             2.608132 0.03819683     0.06431947
61 AIMS061  3.684271             1.922150 0.03814453     0.03579842
40 AIMS040  8.454545             1.910211 0.03442049     0.03179258
46 AIMS046  8.454545             1.910211 0.03442049     0.03179258
5  AIMS005 14.878788            -1.705169 0.03754918     0.02790058
42 AIMS042  8.454545             1.598261 0.03442049     0.02246382
11 AIMS011 14.878788             1.499536 0.03754918     0.02169832
16 AIMS016 14.878788             1.499536 0.03754918     0.02169832
```

```r
quasi_add_no22 <- update(quasi_add, data = reef[-22, ])

quasi_add_no_22_24 <- update(quasi_add, data = reef[-c(22, 24), ])

cbind(Full = coef(quasi_add),
      No_AIMS022 = coef(quasi_add_no22),
      No_AIMS022_024 = coef(quasi_add_no_22_24))
```

```
                           Full  No_AIMS022 No_AIMS022_024
(Intercept)            0.75788860  0.82212517     0.87099217
latitude_bandCentral  -0.01904819 -0.01904819    -0.01904819
latitude_bandSouthern  1.37681562  1.28738509     1.21776710
zoneNo-take            0.56523235  0.46256791     0.38165897
```

```r
summary(quasi_add)
```

```

Call:
glm(formula = coral_trout ~ latitude_band + zone, family = quasipoisson(link = "log"),
    data = reef)

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)           0.75789    0.25166   3.012  0.00319 **
latitude_bandCentral -0.01905    0.32085  -0.059  0.95276
latitude_bandSouthern  1.37682    0.24624   5.591 1.52e-07 ***
zoneNo-take            0.56523    0.17429   3.243  0.00154 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for quasipoisson family taken to be 5.404279)

    Null deviance: 977.60  on 119  degrees of freedom
Residual deviance: 580.03  on 116  degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 5
```

```r
summary(quasi_add_no22)$coefficients
```

```
                       Estimate Std. Error     t value     Pr(>|t|)
(Intercept)           0.82212517  0.2258747  3.63973933 4.102955e-04
```

```
latitude_bandCentral  -0.01904819  0.2887153 -0.06597569 9.475117e-01
latitude_bandSouthern  1.28738509  0.2237255  5.75430551 7.329348e-08
zoneNo-take            0.46256791  0.1608187  2.87633176 4.796071e-03
```

```
summary(quasi_add_no_22_24)$coefficients
```

```
                       Estimate Std. Error     t value     Pr(>|t|)
(Intercept)            0.87099217  0.2109978  4.12796738 6.995429e-05
latitude_bandCentral  -0.01904819  0.2701804 -0.07050176 9.439178e-01
latitude_bandSouthern  1.21776710  0.2111924  5.76615033 7.052381e-08
zoneNo-take            0.38165897  0.1539402  2.47926830 1.462825e-02
```

Removing AIMS022 reduced the No-take coefficient from 0.565 to 0.463, while removing both AIMS022 and AIMS024 reduced it further to 0.382. The Southern coefficient similarly decreased from 1.377 to 1.287 and 1.218. However, both effects remained in the same direction and retained statistical support. These observations therefore affected the magnitude of the estimated effects but did not change the main conclusions, so all sites were retained in the primary analysis.

The substantial extra-Poisson variation had already been accommodated by the quasi-Poisson model. Although a few extreme observations remained, the residual plot showed no clear systematic pattern, and the sensitivity analysis indicated that the main conclusions were robust. Within the models covered in the subject, there was therefore no strong diagnostic evidence that an alternative model was required, so the additive quasi-Poisson model was retained as the final model.

Part 9: Estimates and uncertainty for the client

```
newdat <- expand.grid(latitude_band = levels(reef$latitude_band),
                      zone = levels(reef$zone))

pred <- predict(quasi_add,
                newdata = newdat,
                type = "link",
                se.fit = TRUE)

newdat$fit <- exp(pred$fit)
newdat$lower <- exp(pred$fit - 2 * pred$se.fit)
newdat$upper <- exp(pred$fit + 2 * pred$se.fit)

newdat
```

```
  latitude_band    zone       fit     lower     upper
1      Northern    Open  2.133766  1.289893  3.529717
2       Central    Open  2.093506  1.260652  3.476588
3      Southern    Open  8.454545  6.284222 11.374413
4      Northern No-take  3.755123  2.349495  6.001691
5       Central No-take  3.684271  2.295586  5.913024
6      Southern No-take 14.878788 11.779581 18.793396
```

```
b_zone <- coef(quasi_add)["zoneNo-take"]
se_zone <- sqrt(vcov(quasi_add)["zoneNo-take", "zoneNo-take"])
```

```r
(ci_link <- c(estimate = b_zone,
              lower = b_zone - 2 * se_zone,
              upper = b_zone + 2 * se_zone))
```

```
estimate.zoneNo-take    lower.zoneNo-take    upper.zoneNo-take
           0.5652323            0.2166535            0.9138112
```

```r
(ci_ratio <- exp(ci_link))
```

```
estimate.zoneNo-take    lower.zoneNo-take    upper.zoneNo-take
            1.759857             1.241914             2.493809
```

```r
pred_table <- transform(newdat,
                        fit = round(fit, 2),
                        lower= round(lower, 2),
                        upper = round(upper, 2))

knitr::kable(
  pred_table,
  col.names = c("Latitude band", "Zone", "Fitted mean",
                "Lower 95% CI", "Upper 95% CI"),
  caption = "Fitted site-level mean counts and approximate 95% CIs."
)
```

Fitted site-level mean counts and approximate 95% CIs.

Latitude band	Zone	Fitted mean	Lower 95% CI	Upper 95% CI
Northern	Open	2.13	1.29	3.53
Central	Open	2.09	1.26	3.48
Southern	Open	8.45	6.28	11.37
Northern	No-take	3.76	2.35	6.00
Central	No-take	3.68	2.30	5.91
Southern	No-take	14.88	11.78	18.79

Fitted site-level mean counts from the final additive quasi-Poisson model are shown in the table above. Expected counts were similar in the Northern and Central bands, but substantially higher in the Southern band. The highest fitted mean was for Southern No-take sites, at 14.88 coral trout per site.

Because the final model did not retain the interaction, the estimated No-take/Open comparison is constant across latitude bands. On the linear predictor scale, the estimated zone coefficient was 0.565 with an approximate 95% confidence interval of 0.217 to 0.914. Exponentiating gives an expected-count ratio of 1.76, with a 95% confidence interval of 1.24 to 2.49. Thus, after accounting for latitude band, No-take sites were associated with approximately 76% higher expected coral trout counts than Open sites.

Overall, coral trout abundance was highest in the Southern band, and No-take sites had higher expected counts than Open sites. However, there was insufficient evidence that the association with fishing zone differed among latitude bands after accounting for overdispersion.